

Product Carbon Footprint Study Report

Product Name: textile, knit cotton

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: textile, knit cotton

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg textile, knit cotton

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg textile, knit cotton.

2. System Boundary

textile production, cotton, circular knitting, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of textile, knit cotton produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **6.630 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

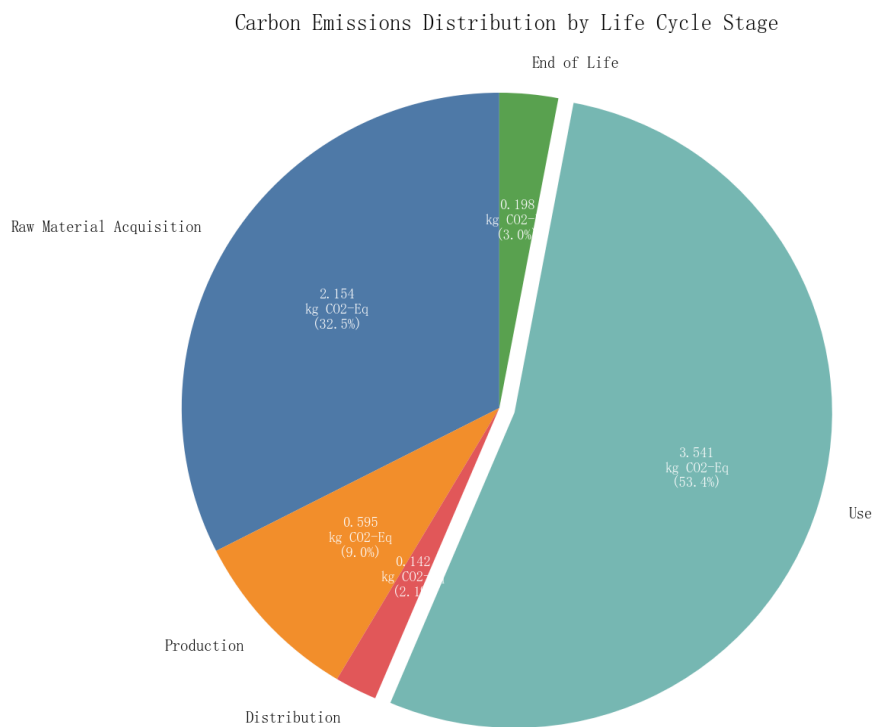


Table 1: Carbon Emissions by Life Cycle Stage

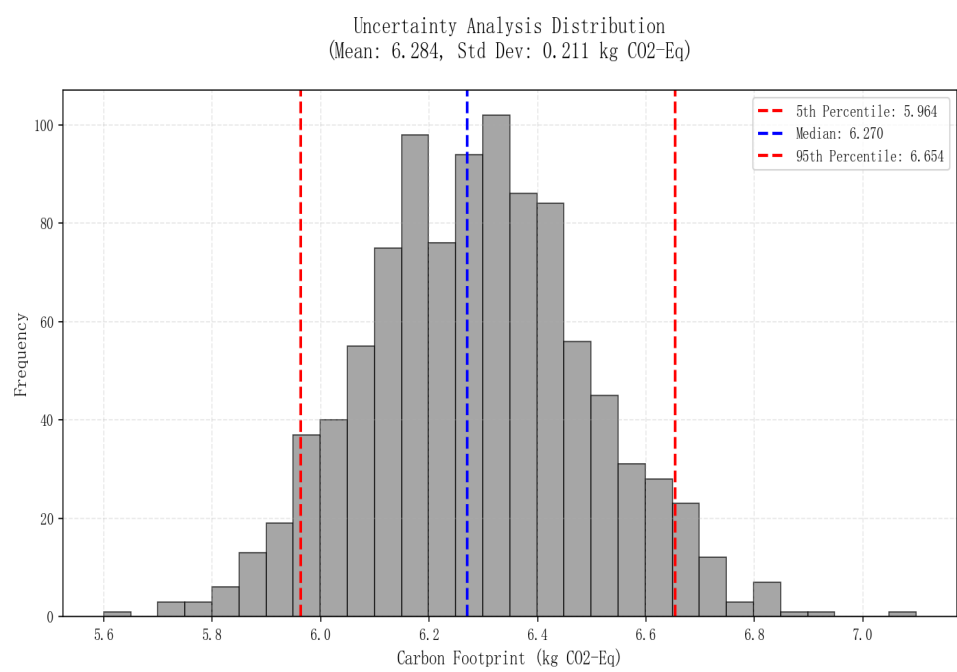
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	2.154	32.48
Production	0.595	8.98
Distribution	0.142	2.14

Use	3.541	53.41
End of Life	0.198	2.98
Total	6.630	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	6.284
Standard Deviation	0.211
Coefficient of Variation (CV)	3.36%
Median	6.270
5th Percentile	5.964
95th Percentile	6.654

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the knit cotton textile product is 6.630 kg CO₂-Eq, representing a moderate emission intensity typical for apparel items in cradle-to-grave assessments. This value reflects the aggregated greenhouse gas emissions across the entire product life cycle, providing a benchmark for environmental performance evaluation and comparison with alternative materials or production methods.

Lifecycle stage contributions reveal a highly skewed distribution, with the use phase accounting for 53.4% (3.541 kg CO₂-Eq) and raw material extraction for 32.5% (2.154 kg CO₂-Eq). These two stages together comprise over 85% of total emissions, while production (9.0%), end-of-life (3.0%), and distribution (2.1%) play minor roles. Such characteristics indicate that consumer behavior and upstream agricultural or fiber processing activities dominate the impact profile.

Major emission sources are therefore concentrated in the use phase, primarily driven by repeated laundering and drying, and in raw material acquisition, linked to cotton cultivation, irrigation, and fertilizer use. Targeting these hotspots offers the greatest potential for meaningful reductions in overall carbon footprint.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a standardized product lifetime with an average of 50 washing cycles at 40°C using European grid electricity, average consumer drying habits, and a functional unit defined as one kilogram of knit cotton fabric. Transportation distances for distribution were modeled using generic European logistics data, and end-of-life scenarios assumed 50% landfill and 50% incineration rates without energy recovery credits.

Limitations of data sources and quality arise from reliance on secondary databases such as Ecoinvent for background processes, which may not fully capture regional variations in cotton farming practices or specific manufacturing efficiencies. Primary data from the producer were unavailable for raw material and production stages, introducing potential uncertainty of ±15-20% in those contributions. Data age and technological representativeness could also affect accuracy for rapidly evolving textile supply chains.

System boundary restrictions encompass a cradle-to-grave scope but exclude indirect land-use change impacts, water scarcity, and toxicity-related categories. Downstream consumer transport to retail and packaging production were omitted, while maintenance activities beyond washing were not included, potentially underestimating real-world emissions.

3. Suggestions for Improvement

- Target the use phase by developing consumer guidance on low-temperature washing (30°C) and air-drying to reduce emissions by up to 40% in this dominant stage.
- Shift raw material sourcing toward certified organic or recycled cotton to lower cultivation-related impacts, potentially cutting the 2.154 kg CO₂-Eq contribution by 25-30%.

- Implement energy-efficient machinery and renewable energy in production facilities to address the 9% production share while collecting site-specific primary data for improved accuracy.
- Extend product durability through design improvements and promote repair services to reduce overall washing frequency and lifetime emissions.
- Enhance data quality by conducting supplier-specific surveys and integrating real-time logistics data to refine distribution and end-of-life modeling.